Environmental Pollution xxx (2017) 1–7

Contents lists available at ScienceDirect

Environmental Pollution

journal homepage: www.elsevier.com/locate/envpol

Effects of air pollution on infant and children respiratory mortality in four large Latin-American cities☆

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ARTICLE INFO

Article history:
Received 31 March 2017
Received in revised form
24 August 2017
Accepted 25 August 2017
Available online xxx

Keywords:
Air pollution
Infant mortality
Children mortality
Respiratory diseases
Time series

ABSTRACT

Objectives: Air pollution is an important public health concern especially for children who are particularly susceptible. Latin America has a large children population, is highly urbanized and levels of pollution are substantially high, making the potential health impact of air pollution quite large. We evaluated the effect of air pollution on children respiratory mortality in four large urban centers: Mexico City, Santiago, Chile, and Sao Paulo and Rio de Janeiro in Brazil.
Methods: Generalized Additive Models in Poisson regression was used to fit daily time-series of mortality due to respiratory diseases in infants and children, and levels of PM_{10} and O_3. Single lag and constrained polynomial distributed lag models were explored. Analyses were carried out per cause for each age group and each city. Fixed- and random-effects meta-analysis was conducted in order to combine the city-specific results in a single summary estimate.
Results: These cities host nearly 43 million people and pollution levels were above the WHO guidelines. For PM_{10} the percentage increase in risk of death due to respiratory diseases in infants in a fixed effect model was 0.47% (0.09–0.85). For respiratory deaths in children 1–5 years old, the increase in risk was 0.58% (0.08–1.08) while a higher effect was observed for lower respiratory infections (LRI) in children 1–14 years old [1.38% (0.91–1.85)]. For O_3, the only summarized estimate statistically significant was for LRI in infants. Analysis by season showed effects of O_3 in the warm season for respiratory diseases in infants, while negative effects were observed for respiratory and LRI deaths in children.
Discussion: We provided comparable mortality impact estimates of air pollutants across these cities and age groups. This information is important because many public policies aimed at preventing the adverse effects of pollution on health consider children as the population group that deserves the highest protection.

1. Introduction

Air pollution has become an important public health concern for most cities in the world. Evidence is mounting that exposure to ambient levels of air pollution has several health impacts ranging from cause-specific mortality and morbidity in adults and the elderly (Brunekreef and Holgate, 2002; Kampa and Castanas, 2008) to impacts in early life (Hajat et al., 2007) and during pregnancy (Dadvand et al., 2013).

Infants and children make up a particularly susceptible population to the health effects of air pollution. Lungs continue to develop after birth and while immature are less able to deal with toxic damages (Dixon, 2002; Pinkerton and Joad, 2000; Schwartz, 2004). In addition, this population is proportionally more exposed than adults to ambient air pollution due to longer periods spent outdoors, higher ventilation rates and mouth breathing, factors that increase their intake of air pollutants (Bateson and Schwartz, 2008; Dixon, 2002; Gilland et al., 1999; Schwartz, 2004; WHO, 2005).

The consequences of exposure to air pollution during early life include impairment of lung function (Harding and Maritz, 2012), increased risk of respiratory illness (Xiao et al., 2016) and a higher probability of premature mortality (Yorifuji et al., 2016; Hajat et al., 2007). Evidence indicates that these impacts continue up to later life and adulthood (Osmond and Baker, 2000) which justifies the

☆ This paper has been recommended for acceptance by David Carpenter.
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https://doi.org/10.1016/j.envpol.2017.08.125
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increasing global concern about exposure to air pollution in this period of life.

However, there are relatively few studies that have examined the effect of air pollution on mortality during childhood, and those who have tried to examine this association may not have had sufficient statistical power due to the small number of events (Bhaskaran et al., 2013), a feature that is common to most cities of developed countries, where infant and child mortality is low (UNICEF, 2014). Therefore, large developing country cities, such as those in the Latin America Region, constitute an ideal scenario to evaluate the mortality effects of pollution for different specific causes in this age group.

Latin America has a very large children population (nearly 9% of the total population in the region or more than 53 million people is under 5 years old), is highly urbanized (over 80% of the population reside in urban settings) (UN-DESA, 2015) and urban levels of air pollution are amongst the higher in the world, making the potential health impact on children related to air pollution quite large.

We evaluated the effect of air pollution on infant and children mortality in four large urban centers in Latin America: Mexico City (21 million habitants) in Mexico, Santiago (5 million habitants), Chile, and Sao Paulo (11 million habitants) and Rio de Janeiro (6 million habitants) in Brazil. By using a common analytical approach, we aimed at providing comparable mortality impact estimates of air pollutants across these cities in these age groups. This information is important because many public policies aimed at preventing the adverse effects of environmental factors on health consider children as the population group that deserves the highest level of protection (WHO, 2005). This study is an extension of the Multicity Study of Air Pollution and Mortality in Latin America (the ESCALA Study) supported by the Health Effects Institute.

2. Methods

A daily time-series study of air pollution and infant and children mortality spanning the years 1997–2005 was conducted in Sao Paulo and Rio de Janeiro, Brazil, Santiago, Chile, and Mexico City, Mexico. This analysis was part of the ESCALA study (Estudio de Salud y Contaminación Atmosférica en Latino America) funded by the Health Effects Institute (Romieu et al., 2012).

Details of the data collected and the analytical approach used can be obtained elsewhere (Romieu et al., 2012). In summary, we examined daily counts of deaths due to respiratory diseases (International Classification of Diseases 10th revision – ICD10 J00-J98) in infants (<1 year old) and children 1–5 years old, and of lower respiratory infections (LRI) ([ICD10] J10-J22) in infants and children 1–14 years old with daily levels of PM_{10} and O_3 in the four cities. For LRI we extended the age group up to 14 years old in order to increase de daily number of events.

We obtained air pollution measurements and mortality data following a standardized protocol and examined the data extensively to ensure comparability among the cities. Daily 8-hr maximum moving average for O_3 and daily 24-hr mean average of PM_{10} were calculated averaging measurements from all monitoring stations in each city. We examined only PM_{10} and O_3 due to data availability and scientific interest. PM_{10} was by far the most common measured pollutant in the cities analyzed, and both PM_{10} and O_3 have been the focus of most studies due to their potential for oxidative stress (Brunekreef and Holgate, 2002). We also considered in the analysis daily mean values of temperature and humidity using natural splines initially with 3 or 6 d.f., each at lags 0, 1, 2, and 3, and then using moving averages.

Following the study protocol agreed upon by researchers of the ESCALA study, we performed the city-specific analysis using Generalized Additive Models (GAM) in Poisson regression to fit the time-series data, according to the equation:

$$\ell n(E(Y_t)) = \beta X_{1t} + \sum_{i=2}^{p} S_i(X_{it})$$

where Y_t and X_{1t} are the number of deaths and levels of air pollution at day t, respectively; X_{it} are the predictor variables, which include time trends and seasonality, and S_i are the smoothing functions using natural splines. Indicator variables for day-of-week and national or local bank holidays were also included to account for the short term cyclic fluctuations in the data.

In the modeling process, we relied on model diagnostics choosing the number of degrees of freedom needed to minimize the Akaike Information Criterion (AIC) and to optimize the Partial Auto Correlation Function (PACF). We also checked the standardized deviance residuals for each meteorological indicator before and after its inclusion in the model.

After building a core model for each city, air pollution levels were introduced in lags of up to three days (single lag models-SLM) and also examining the cumulative effect using constrained polynomial distributed lag models (DLM) (Schwartz, 2000). This later model used a 2-degree polynomial structure with exposure to air pollution consisting of the same day up to lags of 3 days; only the overall effect is reported. Besides considering the latency of the effect of the pollutants, the DLM minimizes the instability in the estimation process, typical of the analyzes that use multiple lags.

The analyses were carried out per cause of death for each age-group. Risk estimates were calculated by introducing the air pollution variables into the models as linear terms and we present them as percentage relative risks for each increment of 10 $\mu g/m^3$ at PM_{10} or O_3 levels assuming a significance level of 5%.

Additional analyses of O_3 concentrations were stratified by season to account for the high seasonal variation of this pollutant. Because of the different geographical position of each city, “warm season” and “cold season” were defined in different ways, with warm season spanning from October to March and cold season from April to September in Sao Paulo, Rio and Santiago, and the opposite for Mexico City.

Finally, fixed- and random-effects meta-analysis was conducted in order to combine the city-specifics results in a single quantitative summary estimate (DerSimonian and Laird, 1986) for each outcome. All analyses were conducted using the software R version 2.15.1 (R Core Team, 2017). A library for R named *ares* was developed for the ESCALA study mostly based on the following R packages: gam, mgcv, stats, splines, and meta.

3. Results

The time series data analyzed spanned a period of 5 years for Rio de Janeiro (2001–2005), 8 years for Sao Paulo (1998–2005) and 9 years for Santiago and Mexico City (1997–2005).

Despite these being the largest cities in Latin America with a considerable children population, the daily number of infants and children deaths by respiratory diseases was low, especially when examining the lower respiratory infections (LRI) subgroup (Table 1). Mexico City presented the highest mean count for respiratory infant mortality (1.8 deaths/day) and Santiago the lowest (0.13 deaths/day).

Mean levels of PM_{10} in these cities varied from 46.7 $\mu g/m^3$ in Sao Paulo to 78.4 $\mu g/m^3$ in Santiago, while for O_3 values ranged from 28.1 $\mu g/m^3$ in Rio to 138.6 $\mu g/m^3$ in Mexico City. As expected, levels of O_3 were higher during the warm season for all cities. Nonetheless, levels of PM_{10} also varied by season being higher during the

Table 1
Summary statistics of daily deaths in São Paulo, Rio de Janeiro, Mexico City and Santiago.

	Resp<1		Resp 1-5		LRI <1		LRI 1-14	
	n	mean (SD)	n	mean (SD)	n	mean (SD)	n	mean (SD)
Sao Paulo	2177	0.75 (0.94)	763	0.26 (0.51)	1807	0.62 (0.86)	761	0.26 (0.51)
Rio de Janeiro	364	0.20 (0.46)	180	0.10 (0.32)	295	0.16 (0.41)	171	0.09 (0.31)
Mexico City	5808	1.80 (1.90)	858	0.30 (0.50)	4838	1.50 (1.60)	783	0.20 (0.50)
Santiago	413	0.13 (0.36)	113	0.03 (0.19)	368	0.11 (0.35)	124	0.04 (0.20)

Resp = respiratory diseases.
LRI = lower respiratory infections.

colder months (data not shown). Mexico City was the driest city and Rio de Janeiro the warmest while the two other cities presented a more temperate weather (Table 2).

The percentage increase in daily mortality for a 10 $\mu g/m^3$ raise in PM_{10} and O_3 levels in single lag models are displayed in Figs. 1–4 for the different combinations of age group and cause of death. There is no discernible lag pattern of effect either for PM_{10} or for O_3. Most of the single lag effects were not statistically significant and for PM_{10} in Sao Paulo and O_3 in Santiago they were negative, indicating a decrease in mortality risk for an increase in pollution levels (except for LRI in 1–14 years-old in Sao Paulo, the only positive effect observed for PM_{10} in that city). Higher effect estimates were observed for Rio de Janeiro but they also exhibited wider confidence intervals.

Overall effects of the distributed lag models (sum of lags 0–3 days) exhibited a similar pattern of results (Table 3). For PM_{10} larger effects were observed for both causes of death in the older age group in Rio de Janeiro while negative and significant results were detected for the younger age group in Sao Paulo. Only LRI in under 1 year old was statistically significant for Mexico City while in Santiago this was the only non-significant result. Fewer significant effect estimates were observed for O_3 noting that for Santiago both results in the older age group were negative. These patterns did not change much when examining the effect of O_3 according to season, although larger effects estimates in the warm season were observed and more results that are negative could be seen in the cold season (Table 4).

Fixed and random effects meta-analysis results are also provided in Tables 3 and 4 for lag 0–3 days. For PM_{10} the percentage increase in risk of death due to respiratory diseases in under 1 year old in a fixed effect model was 0.47% (0.09–0.85). For respiratory deaths in children 1–5 years old the increase in risk was 0.58% (0.08–1.08) and a higher effect was observed for LRI 1–14 years old [1.38% (0.91–1.85)]. No significant results were observed for LRI in under 1 year old.

For O_3, the only summarized effect estimate statistically significant was for LRI in infants. However, the analysis by season showed effects of O_3 in the warm season for respiratory diseases in under 1 year old and negative effects for respiratory deaths in 1–5 years old and LRI in 1–14 years old. Contrasting effects were observed in the cold season (Table 4).

4. Discussion

The four cities analyzed in this study host nearly 43 million people and have a large children population exposed to levels of air pollution well above the recommended WHO guidelines for PM_{10} and O_3. Although the daily number of deaths was relatively low in all cities and results of individual city analysis were quite mixed, our meta-analysis found statistically significant impacts of air pollution on respiratory mortality of infants and children. Fixed effect models for PM_{10} exhibited more consistent results with positive and statistically significant risks for all outcomes except for LRI in infants, a subgroup of respiratory diseases. Fewer positive and significant effect estimates were observed for O_3 and this pattern did not change when examining the effect according to season.

There are not many studies that have thoroughly evaluated the impacts of air pollution on infant or children mortality. Our results are consistent with studies carried out previously in the same cities. In Sao Paulo, Conceição et al. (2001) found that CO, SO_2 and PM_{10} were associated with mortality for respiratory diseases in children under 5 years old. Other studies also in Sao Paulo observed effects of ambient air pollution on neonatal and post-neonatal mortality (Nishioka et al., 2000; Lin et al., 2004). Using a measure of traffic related air pollution, Medeiros et al. (2009) also identified an effect of air pollution in the neonatal period. In Mexico City, exposure 3–5 days before death was associated with increased infant death (Loomis et al., 1999) and a later study confirmed this finding specifically for respiratory-related infant mortality and indicated that infants with lower SES are at higher risk of mortality when exposed to ambient PM_{10} and O_3 (Carbajal-Arroyo et al., 2011).

Other studies have also found a relationship between particulate matter and infant mortality in places with relatively lower pollution levels, such as the United States (Ritz et al., 2006; Woodruff et al., 2006) and Japan (Yorifuji et al., 2016). In this later study, $PM_{2.5}$ was associated with increased risks of infant and post-neonatal mortality related to respiratory causes even when particulate matter concentrations were below Japanese air quality guidelines (Yorifuji et al., 2016). In addition, a meta-analysis of 10 European birth cohorts found an association between air pollution and pneumonia in early childhood (MacIntyre et al., 2014), one of main specific causes of respiratory deaths in children.

Table 2
Summary statistics of air pollutant concentrations and weather conditions in São Paulo, Rio de Janeiro, Mexico City and Santiago.

	PM_{10}		O_3		Temperature		Humidity	
	Mean	(5th, 95th)	Mean	(5th, 95th)	Mean	(5th, 95th)	Mean	(5th, 95th)
Sao Paulo	46.7	(21.5, 89.9)	31.3	(13.2, 56.2)	19.9	(14.4, 24.8)	78.6	(60.4, 94.5)
Rio de Janeiro	36.2	(15.6, 67.1)	13.8)	(2.1, 34.9)	24.6	(19.7, 29.1)	78.7	(67.7, 90)
Mexico City	57.3	(25.0, 101.7)	138.6	(59.6, 215.2)	16.4	(12, 20.6)	49.9	(24.8, 71.4)
Santiago	78.4	(31.9, 149.2)	70.3	(12.9, 123.9)	16.0	(8, 23)	63.0	(38, 88)

Please cite this article in press as: Gouveia, N., et al., Effects of air pollution on infant and children respiratory mortality in four large Latin-American cities, Environmental Pollution (2017), https://doi.org/10.1016/j.envpol.2017.08.125

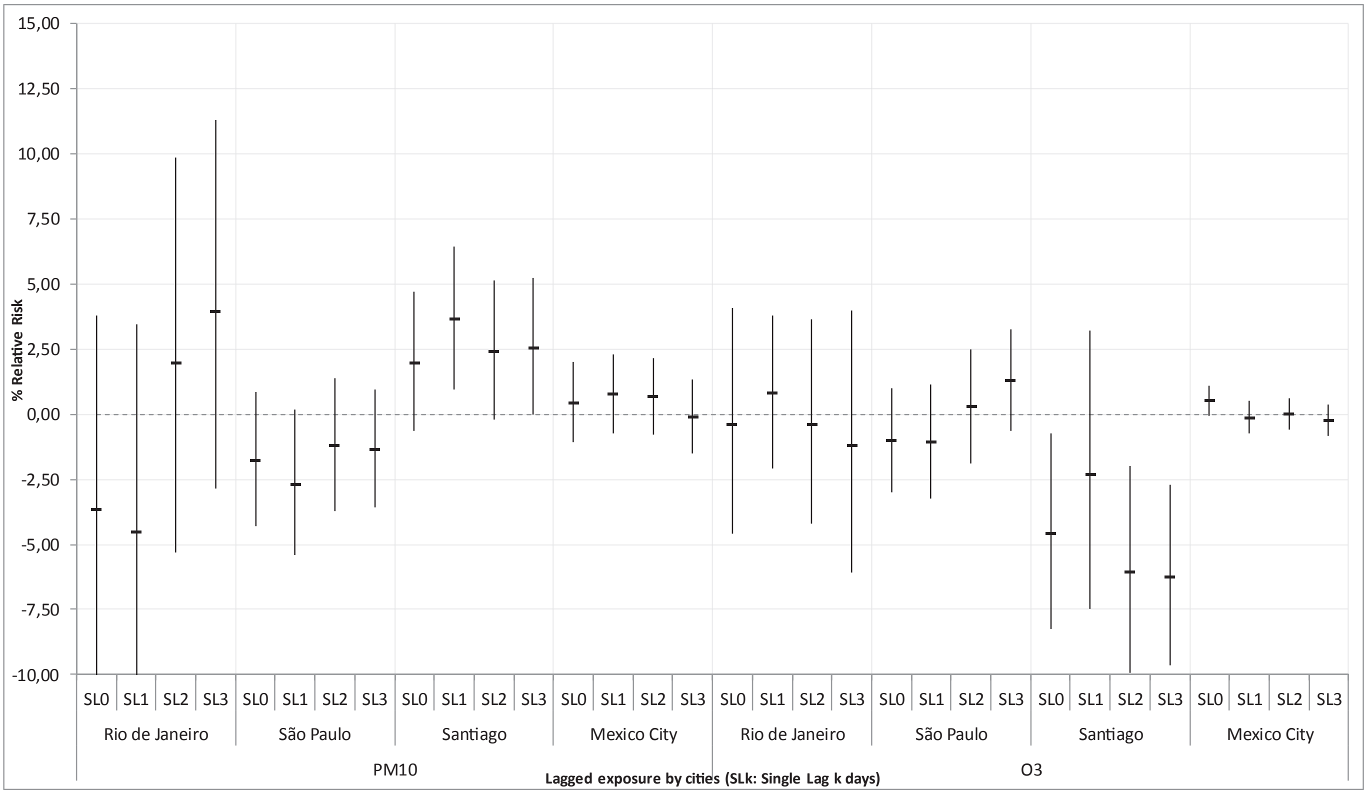

Fig. 1. Percentage increase in daily mortality for respiratory diseases in infants for a 10 μg/m^3 raise in PM_{10} and O_3 levels in single lag models, for Rio de Janeiro, Sao Paulo, Santiago and Mexico City, 1997–2005.

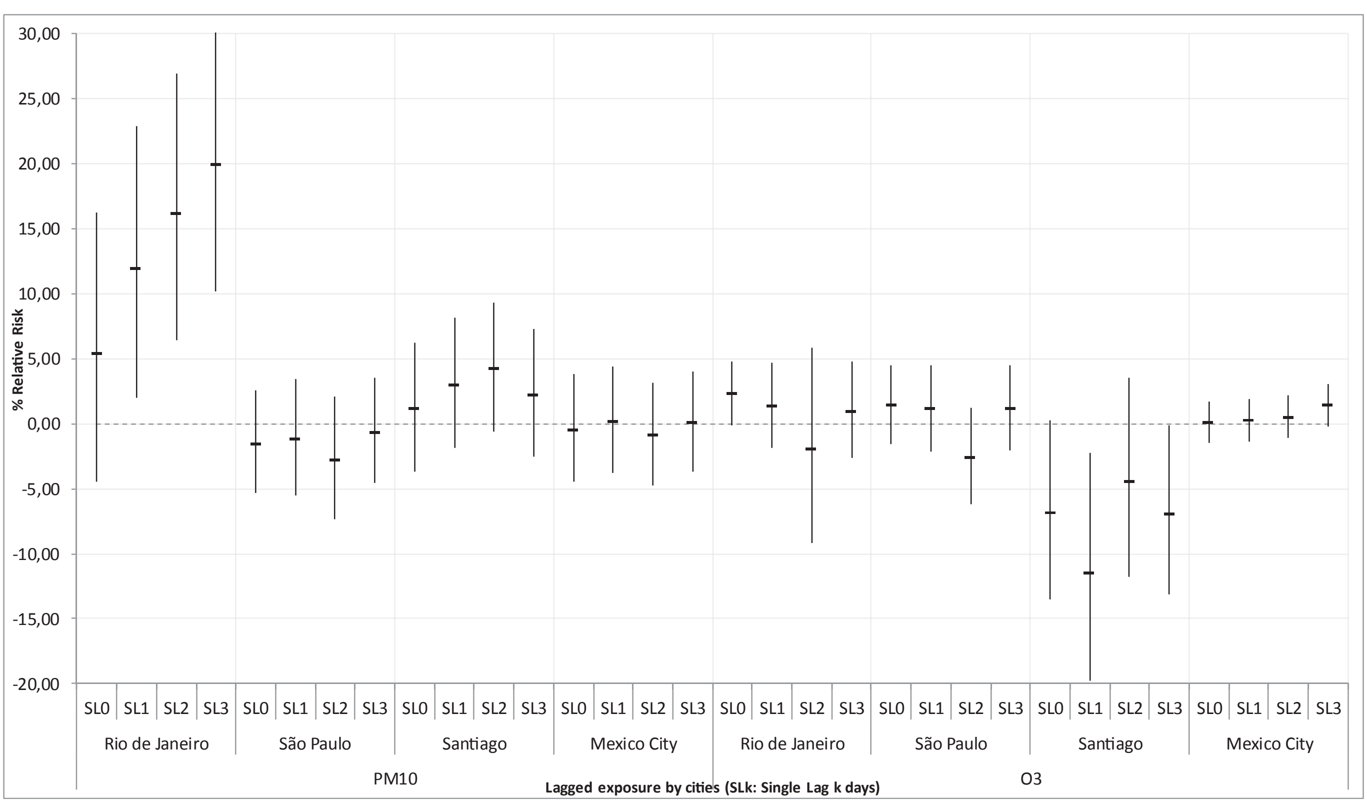

Fig. 2. Percentage increase in daily mortality for respiratory disease in children 1–5 years old for a 10 μg/m^3 raise in PM_{10} and O_3 levels in single lag models, for Rio de Janeiro, Sao Paulo, Santiago and Mexico City, 1997–2005.

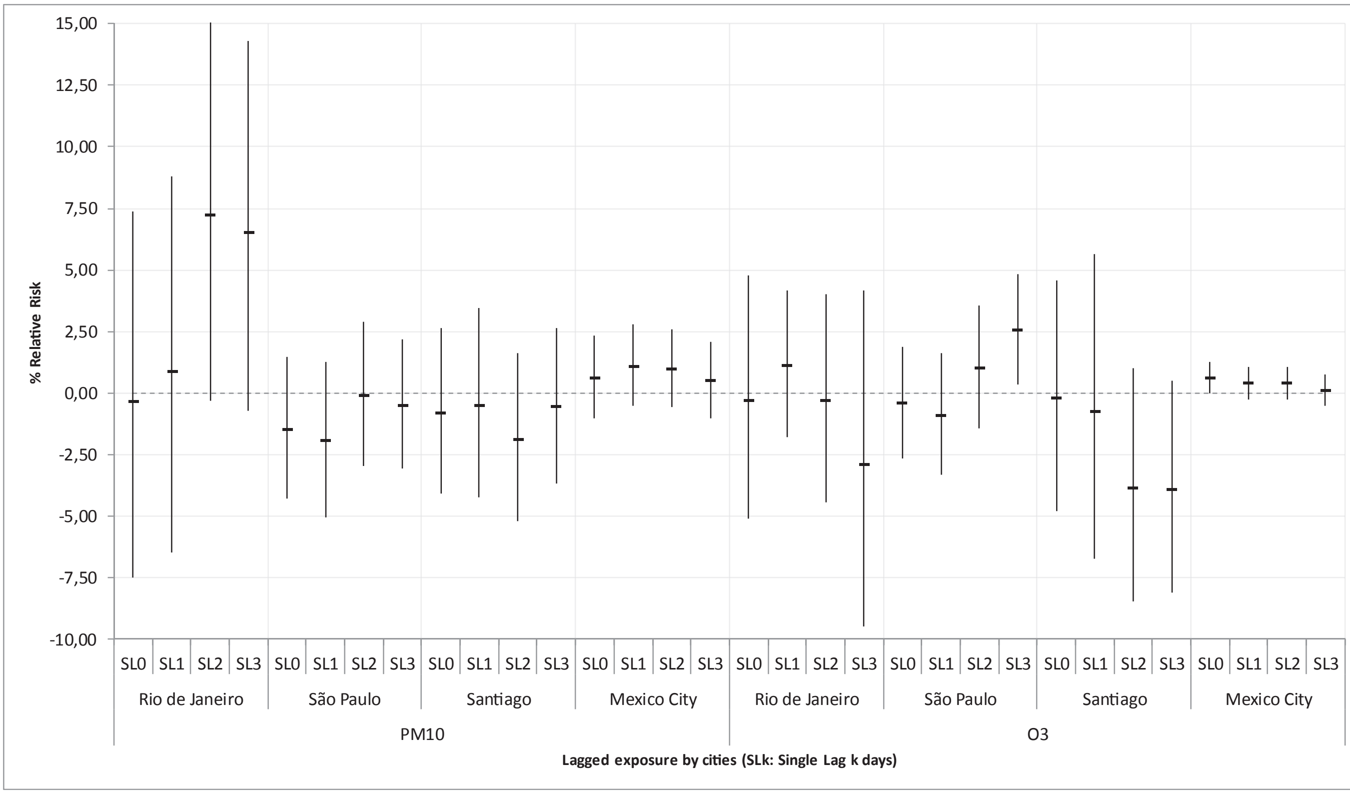

Fig. 3. Percentage increase in daily mortality for acute lower respiratory infections in infants for a 10 μg/m^3 raise in PM_{10} and O_3 levels in single lag models, for Rio de Janeiro, Sao Paulo, Santiago and Mexico City, 1997–2005.

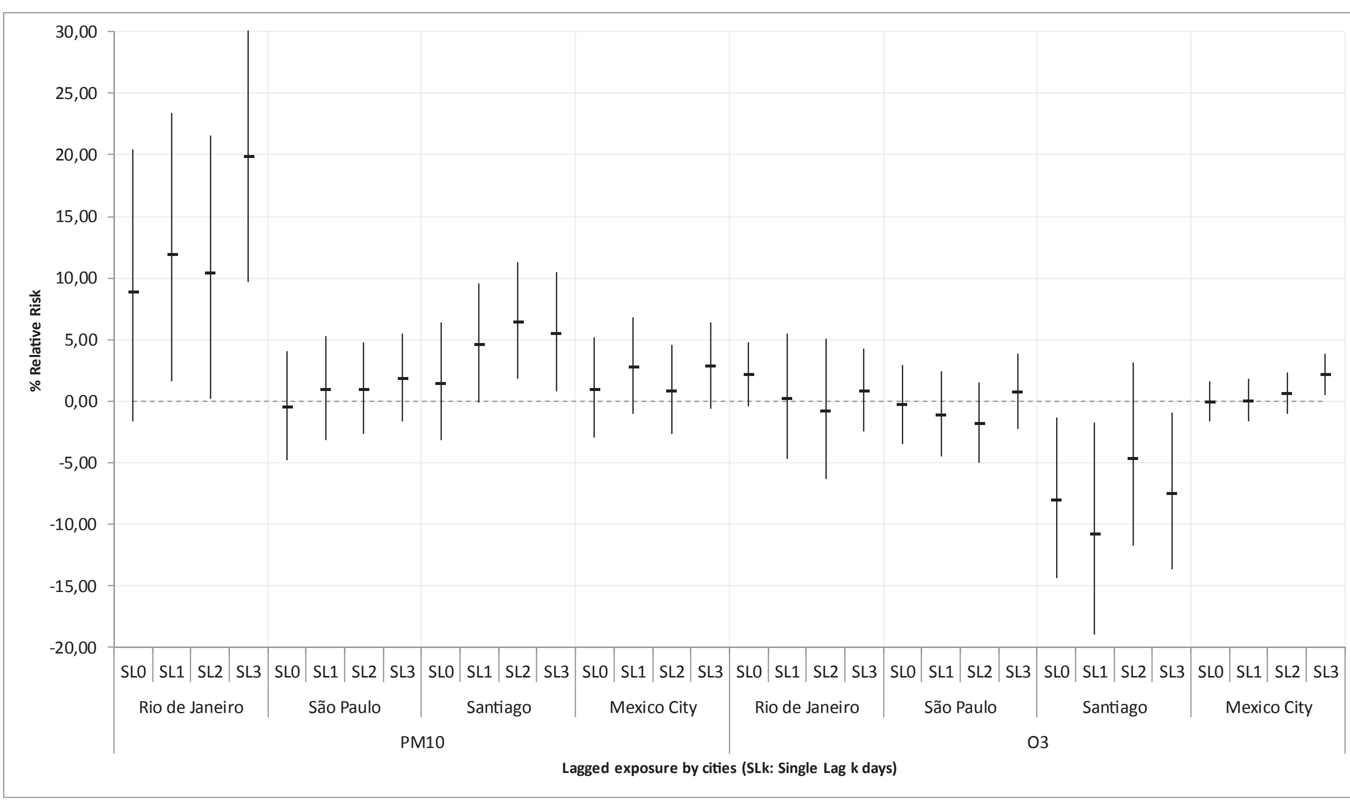

Fig. 4. Percentage increase in daily mortality for acute lower respiratory infections in children 1–14 years old for a 10 μg/m^3 raise in PM_{10} and O_3 levels in single lag models, for Rio de Janeiro, Sao Paulo, Santiago and Mexico City, 1997–2005.

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Table 3
Percentage variation (95% CI) for 10-unit increase in PM_{10} and O_3, by cause of death, in each of the cities. Only overall effects of distributed lag models are presented. Pollutant measure was presented as an average of lags 0–3 days.

		Resp<1		Resp 1-5		LRI<1		LRI 1-14	
		RR[a]	95%CI	RR[a]	95%CI	RR[a]	95%CI	RR[a]	95%CI
PM_{10}	SCL	0.64	(0.23, 1.05)	0.61	(0.10, 1.13)	−1.54	(-3.10, 0.04)	1.28	(0.80, 1.76)
	MC	0.69	(-0.49, 1.88)	−0.24	(-3.42, 3.05)	1.38	(0.09, 2.69)	3.10	(-0.20, 6.51)
	SP	−3.96	(-5.83, −2.05)	−3.03	(-6.01, 0.03)	−2.53	(-4.65, −0.36)	1.40	(-2.05, 4.98)
	RJ	−1.22	(-6.65, 4.53)	21.37	(12.45, 30.99)	5.27	(-0.83, 11.75)	21.56	(12.08, 31.84)
Fixed effect		0.47	(0.09, 0.85)	0.58	(0.08, 1.08)	−0.13	(-1.02, 0.77)	1.38	(0.91, 1.85)
Random effect		−0.70	(-2.57, 1.20)	2.43	(-1.97, 7.03)	−0.16	(-2.54, 2.28)	4.38	(0.36, 8.56)
O_3	SCL	−0.62	(-1.37, 0.13)	−1.50	(-2.45, −0.55)	−0.51	(-1.33, 0.31)	−1.17	(-2.10, −0.23)
	MC	0.11	(-0.33, 0.55)	1.49	(0.31, 2.68)	0.85	(0.37, 1.33)	1.76	(0.54, 3.00)
	SP	−0.26	(-1.73, 1.23)	0.63	(-1.65, 2.95)	1.52	(-0.16, 3.23)	−1.00	(-3.37, 1.41)
	RJ	−0.53	(-3.21, 2.23)	−0.45	(-2.59, 1.74)	−2.00	(-5.02, 1.12)	−1.75	(-4.18, 0.74)
Fixed effect		−0.09	(-0.46, 0.27)	−0.23	(-0.90, 0.44)	0.53	(0.13, 0.93)	−0.27	(-0.95, 0.41)
Random effect		−0.09	(-0.46, 0.27)	0.00	(-1.69, 1.73)	0.27	(-0.82, 1.38)	−0.42	(-2.21, 1.40)

Resp = respiratory diseases.
LRI = lower respiratory infections.
SCL= Santiago, MC = Mexico City, SP= Sao Paulo, RJ = Rio de Janeiro.
[a] Adjusted for trend, seasonality, holidays, temperature, and relative humidity.

Table 4
Percentage variation (95% CI) for 10-unit increase in O_3, by cause of death, in each of the cities for the warm and cold season. Only overall effects of distributed lag models are presented. Pollutant measure is presented as an average of lags 0–3 days.

		Resp<1		Resp 1-5		LRI<1		LRI 1-14	
		RR[a]	95%CI	RR[a]	95%CI	RR[a]	95%CI	RR[a]	95%CI
WARM	SCL	0.71	(-0.74, 2.18)	−5.42	(-7.09, −3.72)	0.45	(-1.28, 2.21)	−3.11	(-4.75, −1.44)
	MC	−0.16	(-1.13, 0.82)	3.31	(0.51, 6.18)	0.27	(-0.86, 1.40)	−0.42	(-2.95, 2.17)
	SP	4.32	(2.36, 6.32)	−3.20	(-8.17, 2.05)	2.24	(-0.15, 4.69)	0.94	(-2.36, 4.37)
	RJ	1.56	(-1.36, 4.58)	−2.35	(-4.72, 0.08)	−1.17	(-4.57, 2.34)	−0.12	(-2.86, 2.70)
Fixed effect		0.80	(0.08, 1.53)	−2.79	(-3.98, −1.57)	0.48	(-0.38, 1.33)	−1.50	(-2.66, −0.33)
Random effect		1.48	(-0.45, 3.45)	−1.96	(-5.95, 2.20)	0.48	(-0.39, 1.36)	−0.98	(-2.91, 0.99)
COLD	SCL	−1.11	(-2.09, −0.13)	0.06	(-1.22, 1.35)	−0.88	(-2.06, 0.32)	−0.42	(-1.64, 0.83)
	MC	0.36	(-0.13, 0.86)	2.02	(0.67, 3.40)	0.78	(0.25, 1.32)	2.53	(1.12, 3.96)
	SP	−7.41	(-9.63, −5.13)	2.23	(-1.95, 6.60)	−6.93	(-9.36, −4.43)	4.73	(0.54, 9.09)
	RJ	−8.75	(-16.95, 0.25)	8.96	(-4.78, 24.67)	−10.29	(-19.00, −0.65)	10.90	(-5.16, 29.68)
Fixed effect		−0.19	(-0.63, 0.24)	1.09	(0.18, 2.01)	0.23	(-0.25, 0.71)	1.10	(0.20, 2.02)
Random effect		−2.93	(-5.63, −0.15)	1.27	(-0.33, 2.89)	−2.70	(-5.62, 0.32)	2.04	(-0.59, 4.75)

Resp = respiratory diseases.
LRI = lower respiratory infections.
SCL= Santiago, MC = Mexico City, SP= Sao Paulo, RJ = Rio de Janeiro.
[a] Adjusted for trend, seasonality, holidays, temperature, and relative humidity.

However, there are studies that could not identify a role of air pollution on children mortality. For example, Hajat et al. (2007) examining 10 British cities found little evidence of the effect of air pollution on infant deaths as few associations were observed with most pollutants studied except Sulphur Dioxide (SO2). However, one should notice that levels of air pollution in these British cities were much lower than what we observed in our four LAC cities, especially for PM_{10}.

The exact mechanisms of how exposure to air pollution affects the respiratory system are not fully understood, but likely involve the interplay of environmental and epigenetic effects (Korten et al., 2017). Studies have suggested that oxidative stress and inflammation are the primary mechanisms by which ambient air pollution induces adverse health effects. These mechanisms may also be relevant to lung function. A meta-analysis of five European birth cohorts examined the association between residential exposure to air pollution and lung function and found that particulate matter was associated with small decreases in lung function in schoolchildren (Gehring et al., 2013) and it is known that impaired lung development contributes to infant mortality. In addition, the effect of air pollution on children respiratory health can initiate even before birth as exposure during pregnancy has also been linked to decreased lung function in infancy and childhood, increased respiratory symptoms, and the development of childhood asthma (Korten et al., 2017).

Like any epidemiological study, this analysis presents some limitations, such as the fact that the daily number of respiratory deaths in all cities was low, which decreases the statistical power of our study for the city-specific analysis. In addition, the fewer negative effects observed for PM_{10} in Sao Paulo and O_3 in Santiago might have been due to chance since there are no biological explanation for a decrease in mortality to be associated with an increase in air pollution. As the single lag analysis showed, there is huge variability in the city specific estimates. Nevertheless, the estimates obtained by the meta-analysis combining data over the four cities should provide a more accurate estimation of the true impact of atmospheric pollution on the children population of these cities. Another limitation is assessing only the impact of air pollution on mortality, a much rarer event compared to morbidity. Although the random-effects estimates are reported, it is more reasonable to acknowledge the fixed-effects ones since all the studies followed a common protocol. Furthermore, given the small number of cities analyzed, the between-cities variance is probably poorly estimated and the fixed-effect model is the appropriate combined measure of association to consider.

Despite these limitations, our results provide further evidence of the effects of air pollution in respiratory mortality of children and infants and are consistent with studies carried out in other large urban centers.

Estimating the global impact of air pollution on the children population of the Latin American region is important to help design interventions that aim to reduce population exposure, as well as to estimate the economic costs of the health effects of this exposure. This information can also support the monitoring of variations in health effects resulting from these possible interventions. As air quality is increasingly highlighted as an important determinant of population health, cities must strive to ensure a better quality of air and, consequently, a better quality of life for its inhabitants. Considering the high levels of air pollution in most large Latin-American cities, the large children population, and the wide socio-economic disparities, it is imperative to take rigorous actions to reduce air pollution exposure and thus long-term respiratory mortality.

Funding

This work was supported by the Health Effects Institute (4745-RFPA04-6/06-6), Boston, MA, USA.

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